Milestone 2

Engineering Design Specification

Bring ROAR to Life - Team 1

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EML 4501 – Senior Design 1

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Executive Summary:

Bring ROAR to Life is a senior design project with the purpose to produce a walking skeletal stegosaurus animatronic with realistic gait. This document will discuss our accomplishments to date, look at competitor benchmarking, and give an overview of our base customer and engineering requirements.

Introduction and Project Goals:

This project originated from Dr. Kurt Stresau, who many years ago purchased for his daughter a stuffed toy of a stegosaurus from a museum gift shop [4]. Several years later Dr. Stresau was inspired by his daughter and "Rex-y" modeled after a complete fossil specimen, AMNH 5027, from the American Museum of Natural History in New York City [1] in the film *Night at the Museum,* to submit a senior design project around the concept. This project will consist of creating a biologically-realistic stegosaurus skeleton, primary focuses will include a biologically accurate appearance, anatomically correct movement, and ease of control.

Figure 01: Image of ROAR Stresau for Reference [4].

When getting familiarized with the project, the team was permitted a high degree of flexibility in determining what direction a final product may look like. Ultimately, the team decided that the target customer would be a museum that would utilize the animatronic as an educational tool/aid within an exhibit or showcase. The primary user will consist of an employee of said museum and will exclude guests or visitors. The secondary user is the companies we need to keep them in mind in terms of cost of maintenance and other aspects of the design because they have a separate need from the operator. The tertiary user is the audience; this is important as the design also needs to reflect and consider the experience from their perspective. The design is not intended to be used for military or industrial purposes and furthermore should not by any means be used to endanger injury or hurt any of the three forms of users.

We have currently met with and interviewed the Lawrence Hall of Science to better understand the wants and needs of our user. We are in active communication with them and will continue to do so throughout the progression of the project.

The primary feature that will stand out in our design will be that it is based on an anatomically accurate skeleton. This is different from what exists in the industry as animatronics to date typically use a “skin” that is stretched or placed over a larger inner structure. Here we are attempting to showcase biological movement, and the skeleton makes for a difficult packaging challenge in order to effectively hide the mechanical and electronic components. Along with replicating a biologically accurate gait.

For our accomplishments to date, we have access to a 3D scan of a stegosaurus skeleton which we can base our design off of. Additionally we are working on a rough 3D printed model and joint design to get rough ideas on the table.

Customer Requirements:

In order to determine what a customer would want the end goal of this project to be, it must first be defined. For this project, our primary customer will be a science museum. And potentially be used as an education tool that can be interactive with visitors and guests. Anatomical Accuracy and lifelike Movement is a must. And the operating environment will be limited to indoor environmental conditions, and more than likely a very smooth/slippery tile floor. To define our specific customer requirements, we have been in contact with the Lawrence Hall of Science at Berkeley. In our talks with them, the above generalizations.

Below is the derived matrix of requirements for customer wants and needs, along with basic design requirements which will be discussed further. An important factor has also been placed on each requirement, this will drive our future design decisions and serve as a guide for future design decisions.

-Individual Customer Questions and Wants:

In our discussions with the Lawrence Hall of Science at Berkeley. The following questions were proposed in the event that this project would possibly become an addition to their museum [10]. A partnership has not been decided at this time, however this gives us a good base to begin determining our customer requirements. These have been discussed at length with the Museum along with our proposed requirements in the table below. Some of the items below have not been fully defined, as the end prototype will drive its use. But a likely use case will be to supplement STEM education. And to have a hands-on way to explain the design process and engineering.

-Technical and operational questions:

- What is the intended use for the Stegosaurus? Is it designed to be a hands-off display, or would it be robust enough to be actively driven by visitors (including young children) on the science center floor?
- How durable are the controller units, and what is the typical range of the signal?
- How is the Stegosaurus powered (rechargeable battery, standard batteries, or plug-in)? What is the runtime on a full charge, and what is the recharge time?
- What maintenance (daily, weekly, yearly) do you anticipate will be required?
- Would you be able to provide the full technical documentation?

-Questions about the project itself:

- What led your group to design and build this remote-controlled Stegosaurus?
- Are there any specific engineering, robotics, or design concepts that the Stegosaurus demonstrates? If there are, would it be possible to have any prepared educational talking points or materials associated with its construction for our facilitation staff?

Customer vs. Design Requirements													
Customer Requirements \ Design Requirements	Importance	Physical Durability	Walking Speed	Turning Radius	Control Range	Safety Features (Kill Switch)	Runtime	Maintainable	Manufacturing Cost	Visual Accuracy	Flexibility/ Articulation	Animation Segments	External/ Internal Sensor System
Easy to operate	7	X			X	X		X			X		X
Walk Like a Dinosaur	9		X		X					X	X		
Anatomically accurate	9		X	X						X	X		
Semi-affordable	1		X						X				
Repairable	3	X						X					
Run-time of 30 minutes	3		X				X						
Compact size/Easy to transport	1	X				X							
Engaging for Audience	9				X		X			X	X	X	

Figure 02: Customer vs. Design Requirement Matrix.

-Locomotion & Aesthetics:

Looking into our customer requirements, the three most important aspects for a successful project will be that it will look and walk like a real life dinosaur. And that it will be engaging for an audience of museum guests. In order to meet these, the final design will need to be based off of an existing 3D-scan of a Stegosaurus skeleton and display a high visual resemblance to that scan. Additionally a large packaging effort must take place in order to effectively hide all main components, motors, joints, etc. Behind or underneath the already existing bone structure. Individual joints should not be limited to be exact copies of what was present in a dinosaur, but an effort should be made to create a biologically appearing joint and overall movement system.

-User Operation:

Another important question brought up by the museum was who will be operating the dinosaur. The main goal of which would be to have guests be able to interact with a finished product. But in order to remain within the scope of this project and for operating a prototype, a museum employee should be the likely operator as they will receive comprehensive training on how to effectively make this prototype move and interact with its environment and guests. In doing so, there will also need to be an operations manual which will be created.

-Run Time:

For a successful exhibit, the end robot will need to last long enough to be operated for the duration of a demonstration along with the ability to charge quickly in order to use the end product multiple times a day. A simple base target for this will be a minimum run time of 30 min, which can be calculated both in testing, and estimating a needed battery capacity from the typical power draws from electrical components.

-Handling and Storage:

When outside of using the end product with guests, there needs to be consideration for both transport, storage, and repair if and when something breaks. As far as handling is concerned our ideal robot shall be able to be carried by a single user without the need for special lifting equipment or processes. To meet this, the goal will be for the robot to weigh under 50lbs as suggested by OSHA [3] in order to help prevent injury.

-Repairability:

If a major component does break, there will need to be readily available instructions on where to source and replicate existing components. It will also be beneficial to use easily replaceable sub-assemblies where possible. This will both speed up the repair process and allow for non-technically trained users to possibly repair a finished device.

-**Audience Engagement:**

One factor that was not previously considered in depth was audience engagement, as the primary end use case for this robot will to be used as a teaching tool for museum guests. This places a greater importance on non-functional movement separate to the primary drive motors and legs. Another aspect to consider is how will this robot be used to teach about STEM outside of being a dinosaur? Which can be translated into using this project in general to teach about the design process and how the flow of a larger engineering project works from conception all the way to a final prototype.

Competitor Benchmarking:

By and large, there is no comparable direct competitor. There are numerous products, however, that contain components of our end goal. For instance, Disney's "Imagineers" have created stationary animatronics with extremely fluid motion [2]. Boston College has developed an animatronic dog, "Spot", that is able to walk semi-autonomously with a realistic gait [3]. None of these companies quite achieve our ultimate goal. They lack the mobility and smooth gait. They are often not anatomically accurate, or have an exterior "skin" hiding the interior skeletal structure. Furthermore, all of these options with the exception of the kids' toys fall far outside the proposed budget. Our hope is to produce a robot that is far closer in price to something that a larger customer base can use while still maintaining anatomical accuracy and natural motion. Institutions who are teaching about dinosaurs like museums would want such primary requirements from an exhibit. The price range will likely be a little outside of the price range of the average consumer. Also, our runtime is noticeably shorter than the other options, mostly because that is the minimum that we expect it to have. In reality, we hope to have a runtime as close to an hour as we can get within the constraints of battery volume and weight. Other than that, our most important requirements are the robot's gait and anatomical accuracy such that the robot is as lifelike as possible. Unlike the toy robot used in our comparison, this robot will not have wheels in order to stabilize its movements, and must be able to dynamically balance on all four legs as it moves like the Boston Dynamics model. This will add accuracy and realism to our own model but poses the largest challenge for our development.

Competitor Requirement Comparison			
Requirements	**Boston Dynamics "Spot"**	**Toy Robot (Vertoy RC velociraptor)**	**ROAR**
Based on a real animal	Yes: Dog	Yes: Velociraptor	Yes
Realistic gait	Yes	No	Yes
Dynamic Balance	Yes	No	Yes
Curved Walking	Yes	Yes	Yes
Remote Control	Yes	Yes	Yes
Run Time	90 mins (avg) 180 mins (Standby)	120 mins	30 mins (minimum)
Static Balance	Yes	Yes	Yes
Anatomical Accuracy	No	Yes	Yes
Walk vs. Run gaits	2 Gaits	1 Gait	2 Gaits
Roar with an open mouth	No	Yes	Yes
stand on hind legs	Yes	No	Yes

Length	43.3"	16.5"	2' - 3' (24" - 36")
Width	19.7"	8.26"	--
Max Height	27.6"	9.44"	--
Net Weight	72.1 lbs	0.634 lb	50 lb max
Max Walking speed	1.6 m/s (3.28 ft/s)	--	0.833 ft/s
Max slope	(+/-) 30 degrees	--	(+/-)10 degrees
Max Step height	11.8"	--	1.5"
Operating Temperature Range	-4 F - 131 F	--	32 F - 100 F
Collision Avoidance Range	13'	--	--
Lighting	>2 lux	--	--
Battery Capacity	564 Wh	--	--
Battery Voltage	--	3.7V	--
Recharge time	60 mins	1 hr	--
Battery Volume	12.8" x 6.6" x 3.7"	1.71" x 2.48" x 0.28"	--
Battery Weight	5.2 kg	0.45g (0.016 oz)	<= 5lb
Breathes steam	Yes	Yes	Undecided
Cost	75,000 USD	49.88 USD	~500 - 1000 USD

Figure 03: Table of Competitor Benchmarks.

Engineering Design Requirements:

In order to understand where the project is heading, it is important to lay out and understand the guiding requirements for this robot. This will set the scope of the project and lead towards the end goal. Below is a copy of Figure 02 for reference.

Customer vs. Design Requirements													
Design Requirements / Customer Requirements	Importance	Physical Durability	Walking Speed	Turning Radius	Control Range	Safety Features (Kill Switch)	Runtime	Maintainable	Manufacturing Cost	Visual Accuracy	Flexibility/ Articulation	Animation Segments	External/ Internal Sensor System
Easy to operate	7	X			X	X		X			X		X
Walk Like a Dinosaur	9		X		X					X	X		
Anatomically accurate	9		X	X						X	X		
Semi-affordable	1		X						X				
Repairable	3	X						X					
Run-time of 30 minutes	3		X				X						
Compact size/Easy to transport	1	X				X							
Engaging for Audience	9				X		X			X	X	X	

Figure 02 (copy): Customer vs Engineering Design Requirements.

The above chart crosses both customer wants and needs to project requirements, this sets a base priority that shall be placed on each requirement. For this project our main focuses will be on realistic walking and speed, articulations, and visual accuracy.

-Visual Aesthetics:

One of the primary goals of this project is for it to simply look like a dinosaur, an end design must be able to meet this and specifically look like a Stegasarus. All designs should be based off of an existing 3-D scan such as the one from the Denver Museum of Nature & Science [9]. Which offers a free 3-D file of their stegosaurus.

To create an exact copy will likely not be realistic for the scope of this project, also the existing 3-D file does not contain elements such as musculature and cartilage. Therefore a degree of freedom shall be permitted in a final design which will be measured off of the base model.

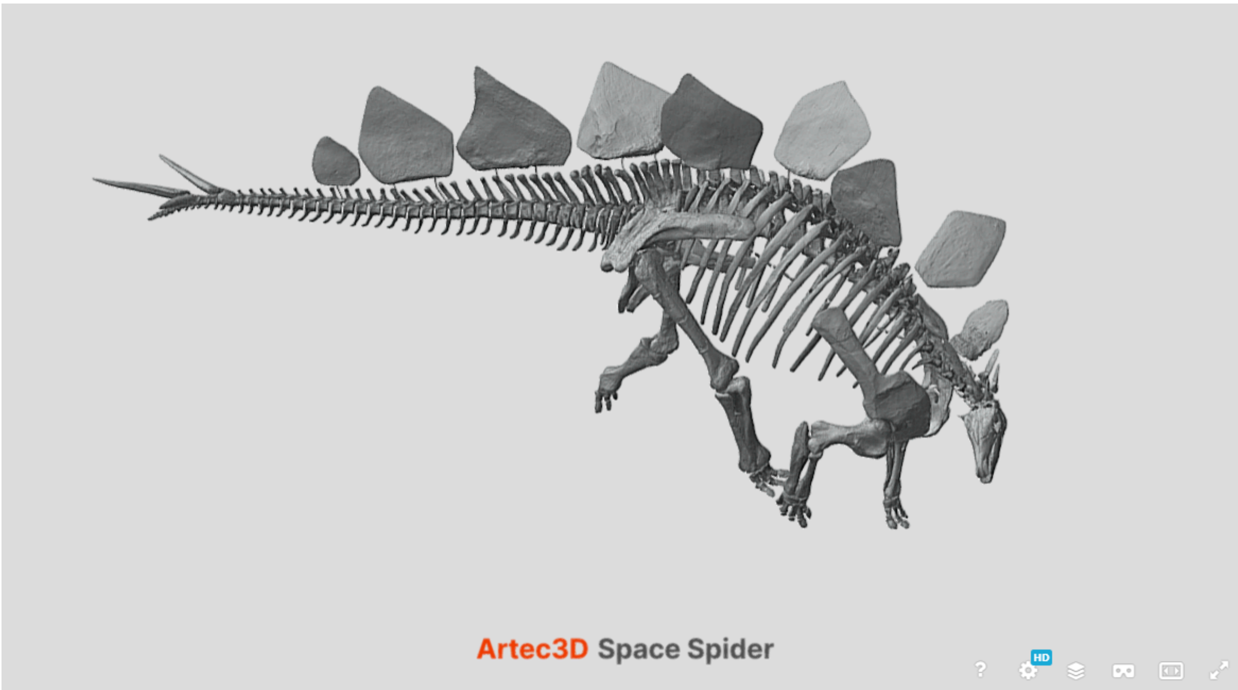

Figure 04: Visual Representation of the Existing 3D Scan [9].

-Walking Speed:

Walking speed and flexibility will make up a large study area for this project. As our primary goal is to emulate realistic movement. In order to make a viable product, it must be able to walk at a reasonable pace.

-Flexibility & Turning Radius:

In addition to having a reasonable walking speed it also must look realistic, that is why it must be flexible enough to emulate a biologically appearing gait. In order to determine this, an extensive study on gait mechanics will be needed in order to determine how far each joint will need to be able to flex. And later on this will be vital on the programming side in order to determine which joints need to move in a timed sequence to create fluid movement.

Along with a realistic walking gait, the robot will also need to be able to turn within a reasonable turning radius. It will be ideal if this value can match that of a stegosaurus, however at minimum it should be able to turn within a radius the same as its own body length from head to tail. This will impact the end joint design in which degrees of freedom will need to be considered.

-Animation Segments:

For a final robot to be interactive with its audience, one consideration will be to add animation segments. Such as a moveable tail, neck, mouth, and so on. This will allow for the robot to display emotion and have an interactive element. These will likely be manually controlled by the end operator. But it may be worth consideration to add on automated elements in order to expand the robots ability to interact with the environment on its own.

-Internal & External Sensors:

A control system will also be needed in order to make the robot move. This will consist of both an internal and external sensor package. Internal sensors shall provide feedback to a main controller, taking user inputs and translating them into motor movement, there shall also be feedback sensors so there is a way to track actual movement vs desired or commanded movement.

There shall also be external sensing, in that the robot should be able to gather data on its external surroundings and environmental factors. Possible data could be but is not limited to, floor tilt in order to balance properly, acceleration forces, displacement from a starting point.

And environmental factors such as temperature. Which can also extend to onboard diagnostics such as motor and electrical component temperatures.

-Control Range:

In addition to the onboard sensor package, there shall also be a way to wirelessly connect the robot to a controller and operator. An extended range is preferred and in a potential exhibit, an operator could remain hidden from a main group of visitors which will increase the engagement factor for an audience. Also it will be nice to operate the robot from a single station and not have to follow the robot closely.

-Runtime:

Runtime will present an important factor in the end usability of the robot, this will set how long potential exhibits can last and the length of time that the robot will be able to interact with guests. As a base target, a finished product shall be able to run for a minimum of 30 minutes, although a longer runtime will be welcomed. In order to meet this, calculations on typical power draws will be run to determine battery sizing. And will interface with onboard sensors in order to determine state of charge and battery health.

-Physical Durability:

In order for a finished product to be used in a museum environment, the end structure must be durable in order to stand up to everyday use. Especially if the robot is to interact with and potentially be operated by guests.

-Maintainence:

In order to make a durable robot, it also must be maintainable and repairable. A regular maintenance schedule should be considered. For the final design, a likely maintenance schedule should consist of periodic cleaning and greasing of joints, charging of the battery, etc. Although the maintenance list should be limited, it will likely be necessary to make a reliable product.

For repairability, swappable sub-assemblies should be used where possible, this will allow for minimal downtime if and when a component breaks. Along with this, a repair manual should be considered that includes general instructions on how to replace these sub-assemblies including needed parts, hardware, tooling, and the like.

-Safety Features:

Safety features will need to be considered especially as this robot will be interacting with members of the public. Some examples of which will be adding a mechanical on-off switch that will disconnect all power to the robot. There shall also be onboard sensing to tell if the battery is getting low on charge and will suspend movement in order to protect the battery. On top of battery limiting and cut offs. There shall also be a way to shut down the robot if it senses that it tips over in order to limit potential damage.

Separate to battery and onboard diagnostics. There shall also be a feature that if the robot loses remote contact with the operator, that it stops. Instead of continuing on its last recorded command. In essence the robot will not be able to continue walking away until it either hits something and potentially injures a guest or damages another piece of its environment.

-Customer vs. Engineering Design Crossmapping:

In creating a relationship matrix, one key aspect is to crossmap the similarities between customer wants and engineering decisions. These drive our primary engineering priorities which for this project include an importance on a realistic walking gait and visual aesthetics.

The similarities within the general requirements can be seen in Figure 02. One customer requirement that we did not anticipate was for a finished product to be engaging to a final audience, this does not greatly impact our final design decisions, however places a greater importance on the ability to display emotion, such as a moveable head, neck, and tail. Along with the ability for the robot to produce sound and have eyes that can light up.

Functional Decomposition:

The following table was made in order to determine our base goals of this project. Primary areas of focus will include Structure, Material, Power, Articulation, and Safety requirements. These are the "shall's" that a final robot should be able to meet. Specific design choices for each system are NOT made at this time, however target values are set which will guide our decisions for each sub-system as the project progresses.

FUNCTIONAL REQUIREMENTS			
Req. #	**REQUIREMENT**	**TARGET**	**VALIDATION METHOD**
S	**STRUCTURAL REQUIREMENTS**		
F.S1	The animatronic shall balance statically.	The animatronic must be able to remain upright on its 4 limbs on a smooth surface within a range of +/- 10° in any cardinal direction.	Test: Stand animatronic on a surface, changing its degree of incline in the cardinal directions.
F.S2	The animatronic shall balance while walking.	Barring any external disturbances, animatronic must retain balance when cycling through all its gaits.	Test: The animatronic will cycle through all gaits on a smooth, flat surface.
F.S3	Joints and structure should support the weight and movement of the robot.	The animatronic shall not weigh more than 50 pounds, including all systems involved. The animatronic shall be able to support its entire weight.	Analysis: G force calculations. FEA.
M	**MATERIAL REQUIREMENTS**		
F.M1	Shall be durable.	Robot shall be able to withstand falls, small drops, and regular	Testing.

		handling.	
F.M2	Shall use additive manufacturing.	3D printing shall be considered for rapid prototyping and custom parts.	Material Analysis and Selection.
F.M3	Shall be economical	Material shall be economical to produce and use standard parts when possible.	Analysis on material costs.
F.M4	Shall be reliable.	Material can consistently withstand the loads and forces applied during operation.	Analysis.
F.M5	Shall be easily repairable.	Shall utilize sub-assemblies that are easily replaceable.	Analysis.
P	**POWER REQUIREMENTS**		
F.P1	Power source shall have a long run time.	Robot runs for at least 30 mins before signaling low battery.	Test.
F.P2	Power systems shall comply with all relevant FCC and Industry safety standards.	Refer to relevant standards.	Analysis, testing.
F.P4	Power distribution infrastructure will not interfere with mechanical articulation.	Joints will be able to have full range of motion without being impeded or damaging the power system.	Test.
F.P5	Power system shall be light weight.	The power system and battery combined must weigh less than 5 lbs.	Analysis.
F.P6	Robot will visibly and audibly communicate diagnostic data.	When the robot reaches 20% battery left, the robot will lay down and its eyes will flash red.	Test.
F.P7	Robot shall be rechargeable.	There will be a USB C port on the back of the robot.	Visual.
F.P8	Software will monitor power source health and fuel consumption.	When the robot reaches below 5% power, it will automatically shut off.	Test.
F.P9	The power system shall use an internal power	the whole robot will function off the same single battery, subsystems	Visual.

	source.	will not have independent power systems.	
F.P10	The power system shall support all current requirements.	The main power source shall be able to handle the max current loads from all internal electronics.	Test.
A	**ARTICULATION REQUIREMENTS**		
F.A1	Shall have a high resolution.	An individual joint should be able to stop and hold within 2 degrees of the desired angle.	Testing.
F.A2	Shall be flexible.	Each leg shall have a minimum of 3 degrees of freedom.	Analysis.
F.A3	Shall emulate realistic movement.	General walking gate and structure will appear to move in a biological way.	Analysis on Gate.
F.A4	Shall have adequate range of motion.	Each joint and limb will move XX degrees to allow for full range of motion.	Analysis.
F.A5	Shall be able to ROAR.	The stegosaurus mouth will open at a 30° angle with respect to the top jaw with a tolerance of +/- 2°.	Test.
F.A6	Shall be able to turn to avoid obstacles.	The animatronic shall be able to turn at a 1:1 body length to turn radius ratio.	Test.
F.A7	Shall be able to move its tail.	Can wag tail at minimum of 45 degrees from the center axis.	Test.
F.A8	Shall be compact.	Wiring and circuitry shall be hidden in the main body of the robot.	Analysis.
C	**CONTROL SYSTEM REQUIREMENTS**		
F.C1	Shall have a control system.	Shall be controlled by a single person.	Test.
F.C2	Shall respond to user input.	Control system shall have a low latency and update rate, with a command latency <= 100ms (Input to Response).	Test.

F.C3	Shall provide and store continuous diagnostics.	Stores Joint positions, IMU, Errors, Response Time, Controller Inputs, Power, and Temperature.	Demonstration.
F.C4	Shall have pre-recorded movement.	The interaction shall fully execute within 10 to 20 seconds then return to its idle state. The animation can be canceled mid execution.	Test.
F.C5	User shall be able to control individual motion.	The controller will provide of head and neck motors in the yaw and pitch directions and hold the position within Plus of minus 2 degrees of the commanded orientation.	Test.
F.C6	Shall make noise/sound.	ROAR volume shall be at minimum 70 decibels.	Test.
F.C7	Movement and walking shall be easily controlled.	The controller shall control direction of gait and speed.	Test.
F.C8	Shall have an operators instruction manual.	Easy to operate and learn control mapping, with an included instruction manual.	Visual.
F.C9	Shall have a balance feedback system.	Stop animation and return to idle if animation is interrupted or loses balance.	Test.
F.C10	Shall be connected wirelessly.	Shall have at least a 50ft range.	Test.
E	**SAFETY AND ENVIRONMENT REQUIREMENTS**		
F.E1	Shall operate in dry conditions.	Robot shall not be submerged in water or exposed to rain.	Test.
F.E2	Shall operate in standard atmospheric conditions.	Standard atmospheric conditions are defined as a temperature of 15°C (59°F) and a pressure of 101,325 pascals (14.7 psi).	Test.
F.E3	Shall have an integrated shut down system.	Program will be able to sense if the robot falls over and turns off.	Test.
F.E4	Shall have a mechanical	Will be able to turn off in less than	Test.

	shut down system.	2 sec if a switch is pressed.	
F.E5	Shall shut down if remote contact is lost.	Shall shut down within 5 seconds of loss of contact.	Demonstration.
F.E6	Shall have low emissions.	Shall not damage its surrounding environment or objects.	Analysis.

Component Decomposition:

The table below contains the individual components we will need in order to build the robot. The components are categorized into Control Systems, Electrical Systems, Main Structure and Joints. The name of each component is listed in the table, and the expected requirement the component must achieve to be viable. Target values for validation are set with their associated units which will guide individual design choices for each component. And should be carefully considered when choosing between various component specifications for a single integrated system.

COMPONENT REQUIREMENTS							
Req. #	**System**	**Component**	**Stretch Goal**	**Requirement**	**Validation Method**	**Units**	**Range**
C.C 1	Control System	On/Off switch		Shall be easily accessible	Inspection	N/A	N/A
C.C 2	Control System	On/Off switch		Shall immediately cut all power to the robot.	Inspection	N/A	N/A
C.C 3	Control System	Remote Transmitter/ Controller		Shall have a minimum range of [50] ft.	Testing	Ft	>= 50
C.C 4	Control System	Remote Transmitter/ Controller		Shall stop all movement if the robot loses range or remote contact.	Testing	N/A	N/A
C.C 5	Control System	Controller		Will contain a computer that is able to be programmed allowing for	Testing	N/A	N/A

				precise control of joints and motion.			
C.C 6	Control System	Motor Controller		Will interface between the primary computer and the motors, combining positioning requests and handling motor electrical requirements.	Testing	N/A	N/A
C.C 7	Control System	IMU	X	Will be able to keep track of inertia when the robot moves.	Testing	N/A	N/A
C.C 8	Control System	Camera	X	Will be able to take in visual data which will affect how the robot moves.	Testing	N/A	N/A
C.C 9	Control System	Gyroscop e	X	Determine whether the position of the robot is upright or if it has fallen over.	Testing	N/A	N/A
C.C 10	Control System	GPS	X	Determine where the robot is located in space.	Testing	N/A	N/A
C.C 11	Control System	I/O		Input/Output connections to support everything and there are enough connections.	Analysis	N/A	N/A
C.E	Electrical System						
C.E1	Electrical System	Battery Pack		Shall have a run time of at least [30] min.	Test	min	>=30
C.E2	Electrical System	Battery Pack		Shall charge in under [180] min.	Analysis	min	<=180
C.E3	Electrical System	Drive Motors		Shall have adequate torque in order to move each limb.	Analysis	N/M	N/A
C.E4	Electrical System	Drive Motors		Shall have adequate speed in order to	Analysis	Rad/Sec	N/A

					replicate a realistic gate.			
C.E5	Electrical System	Animation Motors			Shall have [XX] torque in order to move non-structural elements.	Analysis	N/M	N/A
C.E6	Electrical System	Speaker			Will be able to communicate audio at a volume of at least [XX] decibels.	Testing	dB	>=70
C.E7	Electrical System	Eyes			Will be able to display at least 2 color.	Analysis	Count	>=2
C.S	MAIN STRUCTURE							
C.S1	Main Structure	Power Module			Shall not exceed the physical dimensions of the robot.	Analysis	N/A	N/A
C.S2	Main Structure	Power Module			Shall be able to store and protect all main electrical components with the exception of motors.	inspection	N/A	N/A
C.S3	Main Structure	Power Module			Shall be impact resistant	Inspection	N/A	N/A
C.S4	Main Structure	Hardware/ Fasteners			Shall use off the shelf, metric Nuts & Bolts unless a specific part requires otherwise.	Inspection	N/A	N/A
C.S5	Main Structure	Hardware/ Fasteners			Shall use off the shelf standard size Bearings, Pins, & other hardware.	Inspection	N/A	N/A
C.S6	Main Structure	Bones			Main bone elements will and geometry will be within [XX] percent of measurements taken from an existing scan.	Analysis	Precent	<= 20
C.S7	Main Structure	Back Bone			The main structure, spine & rib cage, shall be able	Analysis	LBS	TBD

				to support [XX] weight.			
C.S11	Main Structure	Fins		Shall have fins along its spine, as the fossil record shows.	Analysis	Count	11
C.S12	Main Structure	Tail Spikes		Shall have tail spikes in accordance to the fossil.	Analysis	Count	4
C.S13	Main Structure	Feet		Interface material shall provide adequate friction to be able to have proper traction.	Analysis	N/A	N/A
C.J	JOINTS						
C.J1	Joints	Leg Joints		Each leg will have at least 3 degrees of freedom.	Analysis	N/A	N/A
C.J2	Joints	Leg Joints		Each joint shall be able to carry the weight of the robot.	Analysis	FOS	>=2
C.J3	Joints	Leg Joints		Each joint shall be able moxe [XX] degrees in order to emulate movement.	Analysis	Degrees	TBD
C.J4	Joints	Neck		Neck will have at minimum [3] moving parts or vertebrae.	Analysis	Count	>=3
C.J5	Joints	Tail		Tail will have at minimum [8] moving parts or vertebrae.	Analysis	Count	>=8
C.J6	Joints	Mouth		Mouth shall be able to open at least [20] degrees.	Analysis	Degrees	>=20

References:

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